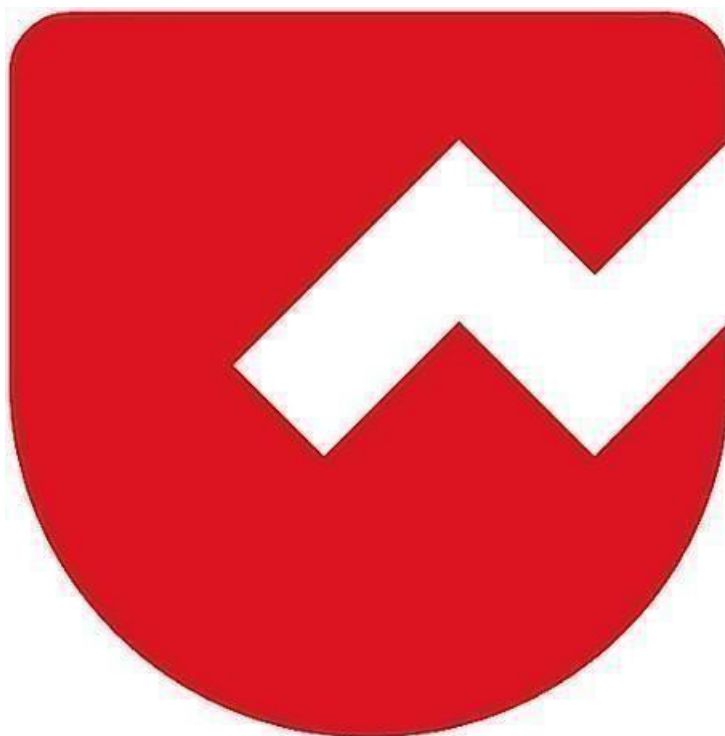


Zoho Schools for Graduate Studies

Notes



Computation of Running Time - Quizzes

Quiz 1

Find the Time complexity:

```
for(int i=1; i<n; i=i*2)  
    System.out.println("Hello");
```

- a) $O(n)$
- b) $O(\sqrt{n})$
- c) $O(\log n)$
- d) $O(n^2)$

Quiz 1 - Answer

Find the Time complexity:

```
for(int i=1; i<n; i=i*2)  
    System.out.println("Hello");
```

- a) $O(n)$
- b) $O(\sqrt{n})$
- c) $O(\log n)$
- d) $O(n^2)$

Quiz 1 - Explanation

Find the Time complexity:

```
for(int i=1; i<n; i=i*2)
    System.out.println("Hello");
```

Explanation:

- 'i' gets incremented as $1(2^0), 2, 2^2, 2^3, \dots, 2^k$
- Suppose if the loop gets terminated after k iterations, then

$$2^k > n$$

$$k > \log(n)$$

- Hence, time complexity = $O(\log n)$

Rules for Algorithmic Analysis

1. Count the number of **basic operations** in each statement (Instruction)
2. Conditional statements : take the longer path
3. Functions : count the statements
4. Loops : count the number of iterations
5. Nested loops : similar to loops
6. Each statement should take constant time
 1. Add, sub, shift, incre/decre : 1 clock cycle
 2. Mul, Div : 1 clock cycle
 3. Memory access : 1 clock cycle
 4. Branch, Return : 3 clock cycle

+

Quiz 2

Find the total basic operations:

```
def fact(n):  
    if n==1:  
        return 1  
    else:  
        return fact(n-1)*n
```

+

- a) $6n+4$
- b) $6n-2$
- c) $9n+4$
- d) $9n-5$

Quiz 2 - Answer

Find the total basic operations:

```
def fact(n):  
    if n==1:  
        return 1  
    else:  
        return fact(n-1)*n
```

+

- a) $6n+4$
- b) $6n-2$
- c) $9n+4$
- d) $9n-5$

Quiz 2 - Explanation

Find the total basic operations:

```
def fact(n):          -> T(n)
    if n==1:          -> 1
        return 1      -> 3
    else:              -> 0
        return fact(n-1)*n -> 3 + 3i + T(n-1) + 1 + 1
```

$T(n) = T(n-1) + 9$, $n > 1$ and $T(1)=4$

Quiz 2- Solving Recurrence Relation

Find the total basic operations:

$$T(n) = T(n-1) + 9, n > 1$$

$$T(n-1) = T(n-2) + 9$$

$$T(n-2) = T(n-3) + 9$$

.....

$$T(2) = T(1) + 9$$

$$T(1) = 4$$

Adding both LHS and RHS, we get

$$\begin{aligned} T(n) &= 9(n-1) + 4 \\ &= 9n - 5 \end{aligned}$$

Quiz 3

Find the recurrence relation:

```
def fibRec(n):  
    if n==0:  
        return 0  
    elif n==1:  
        return 1  
    else:  
        return fibRec(n-1)+fibRec(n-2)
```

Quiz 3 - Options

- a) $T(n) = 15 + T(n-1) + T(n-2), n > 1$
- b) $T(n) = 10 + T(n-1) + T(n-2), n > 1$
- c) $T(n) = 12 + T(n-1) + T(n-2), n > 1$
- d) $T(n) = 14 + T(n-1) + T(n-2), n > 1$

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Quiz 3 – Answer

- a) $T(n) = 15 + T(n-1) + T(n-2), n > 1$
- b) $T(n) = 10 + T(n-1) + T(n-2), n > 1$
- c) $T(n) = 12 + T(n-1) + T(n-2), n > 1$
- d) $T(n) = 14 + T(n-1) + T(n-2), n > 1$

Quiz 3 - Explanation

```
def fibRec(n):           -> T(n)
    if n==0:             -> 1
        return 0         -> 3
    elif n==1 :          -> 1
        return 1         -> 3
    else:
        return fibRec(n-1)+fibRec(n-2)
        -> 3+ (3+1) + T(n-1) + (3+1) + T (n-2) + 1
T(n) = 14 + T(n-1) + T(n-2) , n>1
T(0) = 1 + 3 =4
T(1) = 1 + 1+ 3 =5
```

Quiz 4

Solve the recurrence relation:

$$T(n) = 14 + T(n-1); T(0) = 4; T(1) = 5$$

Quiz 4 - Answer

Solve the recurrence relation:

$$T(0) = 4; T(1) = 5$$

$$T(n) = 14 + T(n-1)$$

$$T(n-1) = 14 + T(n-2)$$

$$T(n-2) = 14 + T(n-3)$$

.....

$$T(2) = 14 + T(1) +$$

$$T(n) = 14 T(n-1) + 5 - O(n)$$

Quiz 5

Find the recurrence relation:

```
def towersOfHanoi(src,tmp,dst,n):  
    if (n>0):  
        towersOfHanoi(src,dst,tmp,n-1)  
        print(src,"to",dst)  
        towersOfHanoi(tmp,src,dst,n-1)
```

- a) $T(n) = 8 + 2T(n-1), n > 0$
- b) $T(n) = 10 + 2T(n-1), n > 0$
- c) $T(n) = 12 + 2T(n-1), n > 0$
- d) $T(n) = 14 + 2T(n-1), n > 0$

Quiz 5 - Answer

Find the recurrence relation:

```
def towersOfHanoi(src,tmp,dst,n):  
    if (n>0):  
        towersOfHanoi(src,dst,tmp,n-1)  
        print(src,"to",dst)  
        towersOfHanoi(tmp,src,dst,n-1)
```

$\rightarrow T(n)$
 $\rightarrow 1$
 $\rightarrow 3+1+T(n-1)$
 $\rightarrow 1$
 $\rightarrow 3+1+T(n-1)$

- a) $T(n) = 8 + 2T(n-1), n > 0$
- b) $T(n) = 10 + 2T(n-1), n > 0$
- c) $T(n) = 12 + 2T(n-1), n > 0$
- d) $T(n) = 14 + 2T(n-1), n > 0$

QUIZ 6

Find the time complexity:

```
for(int i=1; i*i<=n; i++)  
    System.out.println(i);
```

- a) $n\sqrt{n}$
- b) n
- c) $n/2$
- d) $\sqrt{n}$

QUIZ 6 - Answer

Find the time complexity:

```
for(int i=1; i*i<=n; i++)  
    System.out.println(i);
```

- a) $n\sqrt{n}$
- b) n
- c) $n/2$
- d) $\sqrt{n}$

QUIZ 6 - Explanation

Find the time complexity:

```
for(int i=1; i*i<=n; i++)  
    System.out.println(i);
```

Answer: $O(\lfloor \sqrt{n} \rfloor)$ { Floor value of $\sqrt{n}$ }

Explanation:

$$i^2 \leq n$$

$$i \leq \sqrt{n}$$

QUIZ 7

Find the time complexity:

```
for( int i=0 ; i<=n ; i++) {  
    for(int j=1;j<=i;j++) {  
        for(int k=1;k<=100;k++)  
            System.out.println("Hi");  
    }  
}
```

a) $n^2\sqrt{n}$

b) n^3

c) $n^3 \sqrt{n}$

d) n^2

Answer : n^3

QUIZ 8

Find time complexity:

```
While(n>1)  
    n=n/2;
```

- a) $n/2$
- b) $\log_2 n$
- c) n
- d) $\log_2 n - 1$

QUIZ 8 - Answer

Find time complexity:

```
While(n>1)  
    n=n/2;
```

- a) $n/2$
- b) $\log_2 n$
- c) n
- d) $\log_2 n - 1$

QUIZ 8 - Explanation

Explanation:

if $n=2^k$, then, $k=\log_2 n$

Therefore the loop runs $\log_2 n$ times.

QUIZ 9

Find number of basic operations:

```
for(int i=1;i<=m;i++){  
  for(int j=1;j<=n;j++){  
    for(int k=1;k<=p;k++){  
      count++;  
    }  
  }  
}
```

Answer: $3mnp + 4mn + 4m + 2$

QUIZ 9 - Explanation

I	$n=2^k$	$i \leq n$	$n=n/2$	$i++$
0	$2^5=32$	T	16	1
1	$2^4=16$	T	8	2
2	$2^3=8$	T	4	3
3	$2^2=4$	T	2	4
4	$2^1=2$	F		